• Science & Technology

71. Steel slag can be the material for which of the following?

1. Construction of base road
2. Improvement of agricultural soil
3. Production of cement

Select the correct answer using the code given below:

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (d) : Steel slag or by- product of steel-making industries is produced during the separation of the molten steel from impurities in steel making furnaces. The slag occurs as a molten liquid melt and is a complex solution of silicates and oxides that solidifies upon cooling. It is used in various activities such as in road construction, these are the preferred aggregates as they harden well, enabling a thinner layer of pavement than what can be achieved with crushed stone aggregates. Slag is also very hardy, resistant to both weather and wear. So, statement 1 is correct.

Benefits of using steel slag for crops: Slag application favours the increase of pH and the availability of nutrients. Such as Ca, Mg and Si in the soil, which leads to the increase in the absorption of these elements by the plant, favouring the growth and yield of the crops. So statement 2 is correct.

Steel slag can be used to produce energy- saving cement by co-grinding with OPC clinker and blast furnace slag. It can replace Portland cement in various applications and is especially suitable for projects where a low heat of hydration is required. So, statement 3 is correct.

72. In rural road construction, the use of which of the following is preferred for ensuring environmental sustainability or to reduce carbon footprint?

1. Copper slag
2. Cold mix asphalt technology
3. Geotextiles
4. Hot mix asphalt technology
5. Portland cement

Select the correct answer using the code given below:

(a) 1, 2 and 3 only (b) 2, 3 and 4 only
(c) 4 and 5 only (d) 1 and 5 only

Ans. (a) : • The usage of copper slag reduces the usage of primary materials as well as reduces the construction depth, which in turn reduces energy demand in building. So, statement 1 is correct.

• Cold asphalt mix is produced by mixing unheated mineral aggregate with either emulsified bitumen or foamed bitumen. Unlike hot mix asphalt (HMA), Cold asphalt mix does not require any heating of aggregate making, it is economical and relatively pollution free (no objectionable fumes or odours). So, statement 2 is correct and 4 is not correct.

• Geotextile reinforces the soil by adding tensile strength to it. It is used as a rapid dewatering layer in the roadbed. So, statement 3 is correct.

• Portland cement production needs energy and can impact the environment. So, statement 5 is not correct.

73. Consider the following statements:

1. Coal ash contains arsenic, lead and mercury.
2. Coal-fired power plants release sulphur dioxide and oxides of nitrogen into the environment.
3. High ash content is observed in Indian coal.

Which of the statements given above is/are correct?

(a) 1 only (b) 2 and 3 only
(c) 3 only (d) 1, 2 and 3

Ans. (d) : Statement 1 is correct. Coal itself isn't a particularly toxic material. But after it burned, what remains in the ash includes lead, mercury, cadmium, chromium, arsenic and selenium, that may threaten human health.

Statement 2 is correct. Thermal power plants produce a large amount of nitrogen oxides and sulphur dioxide.

• The pollutants that cause acid rain (due to sulfuric and nitric acid) are burning fossil fuels, especially coal to produce energy.

Statement 3 is correct. India's domestic coal reserves have a high ash content - up to 40 to 45 percent.

74. In India, why are some nuclear reactors kept under "IAEA Safeguards" while others are not?

(a) Some use uranium and others use thorium
(b) Some use imported uranium and others use domestic supplies
(c) Some are operated by foreign enterprises and others are operated by domestic enterprises
(d) Some are State-owned and others are privately-owned

Ans. (b) : The nuclear reactors in India are placed under IAEA safeguards only if fuelled by uranium procured from abroad.

At present, 22 operational reactors, of which 14 are under the International Atomic Energy Agency (IAEA) safeguards as these use imported fuel. India currently imports uranium from Russia, Kazakhstan and Canada. Plans are also afoot to procure the fuel from Uzbekistan and Australia. By placing the reactors under the IAEA safeguards, India gives the international nuclear energy watchdog access to them. This step was taken by the country in 2014 to demonstrate that its nuclear energy programme was for peaceful purposes. It was a necessary step under the Indo-US nuclear deal. Hence option (b) is the correct answer.

75. In India, under cyber insurance for individuals, which of the following benefits are generally covered, in addition to payment for the loss of funds and other benefits?

1. Cost of restoration of the computer system in case of malware disrupting access to one's computer.
2. Cost of a new computer if some miscreant willfully damages it, if proved so
3. Cost of hiring a specialized consultant to minimize the loss in case of cyber extortion
4. Cost of defence in the Court of Law if any third party files a suit

Select the correct answer using the code given below:

(a) 1,2 and 4 only (b) 1,3 and 4 only
(c) 2 and 3 only (d) 1, 2, 3 and 4

Ans. (b) : Cyber insurance is designed to guard businesses from the potential effects of cyber attacks. It helps an organisation mitigate risk exposure by of setting costs, after a cyber attack/breach has happened. To simplify, cyber insurance is designed to cover the fees, expenses and legal costs associated with cyber breaches that occur after an organisation has been hacked or from theft or loss of client/employee information. In India, cyber insurance covers (generally) the following:

Identify theft, Cyber-bullying and cyber stalking, cyber extortion. Malware Intrusion: Financial loss due to unauthorised and fraudulent use of bank account, credit card and mobile wallets. Legal expenses arising out of any covered risk, Social media cover, Phishing cover,

E-mail spoofing, Media liability claims cover, Cyber Extortion cover, Privacy breach and data breach by third party.

Hence only option 1, 3 and 4 are correct.

76. With reference to carbon nanotubes, consider the following statements:

1. They can be used as carriers of drugs and antigens in the human body.
2. They can be made into artificial blood capillaries for an injured part of human body.
3. They can be used in biochemical sensors.
4. Carbon nanotubes are biodegradable.

Which of the statements given above are correct?

(a) 1 and 2 only (b) 2, 3 and 4 only
(c) 1, 3 and 4 only (d) 1, 2, 3 and 4

Ans. (c) : Carbon nanotubes (CNTs) are cylindrical molecules that consist of rolled-up sheets of single-layer carbon atoms (graphene). Carbon nanotubes (CNTs) can be functionalised with by active peptides, proteins, nucleic acids and drugs. They are also used to deliver their cargos to cells and organs. Therefore, statement 1 is correct.

The other applications of CNTs have been extensively performed not only for drug and gene therapies but also for tissue regeneration, biosensor diagnosis, enantiomer separation of chiral drugs, extraction and analysis of drugs and pollutants. Hence, statement 3 are true.

CNTs are biodegradable. Several types of microbes including bacteria and fungi can degrade CNTs, graphene and their derivatives. Therefore, statement 4 is also correct.

Hence Statement 1,3 and 4 are correct.

77. Consider the following activities:

1. Spraying pesticides on a crop field
2. Inspecting the craters of active volcanoes
3. Collecting breath samples from spouting whales for DNA analysis

At the present level of technology, which of the above activities can be successfully carried out by using drones?

(a) 1 and 2 only (b) 2 and 3 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (d) : Drones are already well known to spread pesticides and to inspect volcanoes remotely. Hence statements 1 and 2 are true.

Since 2015 researchers at the woods hole oceanographic institute in cape cod have begun using a helicopter-style drone to monitor humpback whales off the coast, collecting breathe samples from their blowholes and taking aerial pictures. Getting a breath sample from a whale isn't the easiest task in the world. But with the help of a hexacopter, scientists are getting the job done. Therefore, statement 3 is also correct.

78. "The experiment will employ a trio of spacecraft flying in formation in the shape of an equilateral triangle that has sides one million kilometers long, with lasers shining between the craft." The experiment in questions refers to -

(a) Voyager-2 (b) New Horizons
(c) LISA Pathfinder (d) Evolved LISA

Ans. (d) : The evolved Laser Interferometer Space Antenna (eLISA) a mission planned that uses three separate, but not independent, interferometers installed in three spacecraft located at the corners of a triangle with sides of same 5 million km (3 million miles).

This mission is to be carried out jointly by NASA and ESA. A mission to test the technology of eLISA with LISA pathfinder, was launched by ESA in 2015. This is